

# Recommendations — hgsoc-tuboovarian-stage3c-r0-hrd-pen ding-idyq

*Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references*

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# Recommendations — hgsoc-tuboovarian-stage3c-r0-hrd-pending

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## Profile snapshot

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- **Primary site:** ovary
- **Histology:** high-grade serous carcinoma (tubo-ovarian / primary peritoneal; WT1+, PAX8+, SOX17+, CK7+)
- **Stage:** IIIc
- **ECOG:** 1
- **Age band:** 70-79
- **Sex:** female
- **Biomarkers:** TP53 (p53 IHC surrogate) mutated (inferred) — p53 diffuse strong nuclear overexpression in >80% of tumor cells; BRCA1/2 (germline) wild-type — no pathogenic or likely-pathogenic variant; HRD / genomic instability (tumor) unknown — not yet resulted (ngs\_pending); Somatic BRCA1/2 (tumor) unknown — not yet resulted (ngs\_pending); MSI / MMR unknown — not yet resulted (ngs\_pending); ER positive (~80% of tumor nuclei, weak-moderate intensity); PR negative; HER2 negative (IHC 1+); FOLR1 (folate receptor alpha) negative (<75% of cells at moderate/strong membranous intensity); CA-125 normalized on therapy (peak 940 -> 11.6 U/mL); Signatera ctDNA (MRD) cleared to Not Detected (30.3 -> 3.32 -> 0 MTM/mL)
- **Efficacy/toxicity weight:** 0.7
- **Toxicity vetoes:** additional peripheral neuropathy, severe myelosuppression / febrile neutropenia
- **Modality constraints:** oral or low-infusion-burden preferred, caution with agents carrying thrombotic or bleeding risk (history of high VTE risk; prior IVC filter)

6 ranked therapeutic options across 4 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

## Hrd Status interventions

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### 1. niraparib (1L maintenance)

**Likelihood of effect:** Moderate across all-comers (PRIMA ITT PFS HR 0.62); high in the HRD-positive subset (HR 0.43). Does not wait on the pending score.

**Toxicity burden:** High (anemia 31%, thrombocytopenia 28.7%, neutropenia 12.8%; cumulative G3+ thrombocytopenia 39.7% at 3.5 yr)

**Counter-productive MoA:** **Moderate** (Cytopenia-driven dose interruptions can erode maintenance dose intensity; the myelosuppression that engages her veto is itself the limiter)

**Overall:** The PARP option that does not wait on the HRD score; all-comers benefit is real but the cytopenia profile sits on her myelosuppression veto and requires the individualized starting dose.

**Key references:** PMID 31562799 · NCT02655016 · PMID 30647846

### 2. olaparib + bevacizumab (1L maintenance) \*(conditional on hrd\_status positive)\*

**Likelihood of effect:** High in the HRD-positive subset (PAOLA-1 HR 0.33; 5-yr OS 66% vs 48%); the largest effect on the board. Foreclosed if HRD-negative.

**Toxicity burden:** Moderate (hypertension 19%, anemia 17% vs <1% on bev alone, lymphopenia 7%; compounded bleeding/VTE on a prior-IVC-filter patient)

**Counter-productive MoA:** **Moderate** (Anti-angiogenic plus PARP stack raises the bleeding/VTE vector that her IVC-filter history already flags; the conservative dissent rested on this mechanism)

**Overall:** The largest-magnitude option when HRD reads positive - but only with a documented bleeding/VTE and CBC monitoring plan, because it stacks PARP onto ongoing anti-VEGF therapy in a high-VTE patient.

**Key references:** PMID 31851799 · NCT02477644

### 3. olaparib monotherapy (1L maintenance) \*(conditional on hrd\_status positive)\*

**Likelihood of effect:** High if somatic-BRCA-mutated (SOLO1 PFS HR 0.30, mPFS 56 vs 13.8 mo; 7-yr OS HR 0.55). Foreclosed if the somatic call is wild-type.

**Toxicity burden:** Moderate (anemia 22%, neutropenia 8%, AE discontinuation 11.5%; long-term AML 1.2%)

**Counter-productive MoA:** **Low** (Cumulative myelosuppression and the long-term myeloid (AML) risk press on her marrow veto across a multi-year maintenance window)

**Overall:** Highest-quality evidence and the only mature OS readout, but the strictest gate - reachable only if tumor NGS returns a somatic BRCA1/2 mutation.

**Key references:** PMID 30345884 · NCT01844986

## Evidence in detail

##### niraparib (1L maintenance)

**\*\*González-Martín et al. *N Engl J Med* 2019\*\***

- **Disease context:** newly-diagnosed advanced ovarian cancer at high risk of recurrence, in response after platinum-based chemotherapy (maintenance) — all-comers regardless of BRCA or HRD status; HRD-positive subgroup pre-specified. Stage III with residual disease or any stage IV. Benefit larger in HRD-positive but present across the ITT, so this option does not strictly require the pending tumor HRD result.
- **n:** 733
- **Toxicities:** anemia (grade ≥3): 31.0% (n=484); thrombocytopenia (grade ≥3): 28.7% (n=484); neutropenia (grade ≥3): 12.8% (n=484)
- **Efficacy:** HR\_PFS 0.62 HR (95% CI 0.5-0.76); HR\_PFS 0.43 HR (95% CI 0.31-0.59); PFS 13.8 months

##### olaparib + bevacizumab (1L maintenance)

**\*\*Ray-Coquard et al. *N Engl J Med* 2019\*\***

- **Disease context:** newly-diagnosed advanced high-grade serous/endometrioid ovarian cancer in

**response after first-line platinum plus bevacizumab** (maintenance) — all on bevacizumab; benefit concentrated in HRD-positive tumours (with or without BRCA mutation). She is already on bevacizumab, so this is the closest design match, but the olaparib add-on only pays off if the pending HRD score is positive.

- **n:** 806
- **Toxicities:**hypertension (grade  $\geq 3$ ): 19.0% (n=535); **anemia** (grade  $\geq 3$ ): 17.0% (n=535); **lymphopenia** (grade  $\geq 3$ ): 7.0% (n=535); **neutropenia** (grade  $\geq 3$ ): 6.0% (n=535); **fatigue/asthenia** (grade  $\geq 3$ ): 5.0% (n=535)
- **Efficacy:**HR\_PFS 0.59 HR (95% CI 0.49-0.72); **HR\_PFS** 0.33 HR (95% CI 0.25-0.45); **OS\_rate** 66.0%; **PFS\_rate** 46.0%

##### olaparib monotherapy (1L maintenance)

**\*\*Moore et al. *N Engl J Med* 2018\*\***

- **Disease context:**newly-diagnosed advanced high-grade serous/endometrioid ovarian cancer with a **BRCA mutation, in response after platinum-based chemotherapy** (maintenance) — germline or somatic BRCA1/2 mutation required; FIGO stage III-IV; complete or partial response to first-line platinum. Excludes BRCA-wild-type tumours, which is the gap for this germline-wild-type, somatic-BRCA-pending patient.
- **n:** 391
- **Toxicities:**anemia (grade  $\geq 3$ ): 22.0% (n=260); **neutropenia** (grade  $\geq 3$ ): 8.0% (n=260); **AML** (grade 5): 1.2% (n=260); **treatment discontinuation for AE** (grade any): 11.5% (n=260)
- **Efficacy:**HR\_PFS 0.3 HR (95% CI 0.23-0.41); **PFS** 56.0 months; **HR\_OS** 0.55 HR (95% CI 0.4-0.76)

## Vegf Angiogenesis interventions

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### 1. bevacizumab (concurrent + maintenance)

**Likelihood of effect:** Moderate for PFS (GOG-218 HR 0.717; ICON7 HR 0.81); OS gain confined to the high-risk subgroup she matched pre-resection. No ITT OS benefit.

**Toxicity burden:** Low (hypertension ~22.9% G2+; bleeding/thrombotic events on a VTE-risk patient; no marrow or neuropathic signal)

**Counter-productive MoA:**Low (Anti-angiogenic wound-healing impairment and the bleeding/VTE vector matter given her IVC-filter history, but the axis is already part of her tolerated regimen)

**Overall:**Lowest-regret guideline-aligned interim that clears both vetoes and holds the VEGF axis steady; modest PFS-only magnitude with no ITT OS gain.

**Key references:** PMID 22204724 · PMID 22204725 · NCT00262847 · NCT00483782

### Evidence in detail

##### bevacizumab (concurrent + maintenance)

**\*\*Burger et al. *N Engl J Med* 2011\*\***

- **Disease context:**newly-diagnosed advanced epithelial ovarian / fallopian-tube / primary-peritoneal

**cancer (stage III incompletely resected or stage IV) (1L)** — all-comers; the PFS benefit came only from the bevacizumab-throughout (concurrent + maintenance) arm, matching her current regimen. This is the direct evidence base for the anti-angiogenic axis she is already on.

- **n:** 1873
- **Toxicities:**hypertension (grade  $\geq 2$ ): 22.9% (n=608); **gastrointestinal perforation/fistula** (grade any): (n=608)
- **Efficacy:**HR\_PFS 0.717 HR (95% CI 0.625-0.824); **PFS** 14.1 months; **HR\_OS**

**\*\*Perren et al. *N Engl J Med* 2011\*\***

- **Disease context:**newly-diagnosed ovarian cancer including high-risk stage IIIC/IV (1L) — all-comers; largest benefit in the high-risk subgroup (stage IV or stage III with  $>1$  cm residual disease), which overlaps her presentation before R0 resection.
- **n:** 1528
- **Toxicities:**hypertension (grade  $\geq 2$ ): 18.0% (n=745)
- **Efficacy:**HR\_PFS 0.81 HR (95% CI 0.7-0.94); **OS** 39.3 months

## Er Expression interventions

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### 1. letrozole (endocrine maintenance)

**Likelihood of effect:** Low-to-moderate (60% vs 38.5% recurrence-free at 24 mo, single non-randomized cohort, pmid:29157627); PR-negativity tempers it further.

**Toxicity burden:** Low (class effects: hot flushes, arthralgia, fatigue; no marrow or neuropathic toxicity)

**Counter-productive MoA:**Low (Cytostatic / antiangiogenic rather than cytotoxic; magnitude tempered by PR-negativity, so the holding effect may be shallow)

**Overall:**The one option that clears every preference axis, but on tier-4 evidence in a PR-negative tumor - a low-toxicity holding strategy, not a frontline backbone.

**Key references:** PMID 29157627 · PMID 24410765

### Evidence in detail

##### letrozole (endocrine maintenance)

**\*\*Heinzelmann-Schwarz et al. *Gynecol Oncol* 2018\*\***

- **Disease context:**estrogen-receptor-positive high-grade serous ovarian cancer, maintenance after first-line therapy (maintenance) — ER-positive high-grade serous histology, the patient's exact biology. Low-burden oral therapy fitting her age band and stated preference; PR-negativity tempers expected magnitude.
- **n:** 80
- **Toxicities:** Aromatase inhibitors are low-toxicity; class effects are hot flushes, arthralgia and fatigue. No high-grade hematologic or neuropathic toxicity, which fits her vetoes. Per-term AE breakdown not extractable from the cohort report.
- **Efficacy:**PFS\_rate 60.0%

## II12 Microenvironment interventions

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### 1. GEN-1 (IMNN-001) program incl. OVATION-3 maintenance trial

**Likelihood of effect:** Unestablished. OVATION-2 topline OS HR 0.74 with no peer-reviewed primary paper; the BRCaM/HRD subgroup CI (0.11-1.70) crosses 1 on n=34.

**Toxicity burden:** Low (intraperitoneal IL-12 manageable in prior GEN-1 work; no structured per-term safety table public; no marrow or neuropathy signal)

**Counter-productive MoA: High** (The critic veto stands on evidence-quality grounds: an unpublished result with a subgroup CI crossing 1 cannot anchor a treatment recommendation)

**Overall: Considered and not recommended: targets her on-study IL-12 axis but rests on unpublished conference topline the critic vetoed - the on-study program is the trial's and treating team's conversation.**

**Key references:** NCT03393884 · NCT06915025